



Git Init

Introduction

Imagine you are starting a **new coding project**. You want to **track all changes**, **collaborate with others**, and **restore previous versions** if needed.

How do you start tracking your project using Git?

This is where the **git init** command comes in!

Section 1: Learn

What is **git init?**

git init is the command that initializes a new Git repository. It tells Git to start tracking the files in your project.

Why Do We Need **git init?**

Without Git (git init)	With Git (git init)
Changes are not tracked	Every change is saved with history
No version history	Can restore previous versions
Difficult to collaborate	Enables team collaboration
No backup of files	Stores code securely on GitHub



How Does **git init** Work?

1. Creates a **hidden .git folder** in your project directory.
2. Starts **tracking all changes** in the project.
3. Allows you to **add, commit, and push code** to a remote repository.

An Interesting Anecdote

Before Git, developers used **manual file backups** (**project_final.doc**, **project_final_v2.doc**). This caused **confusion and lost work**. With Git, everything is **automated and structured**, thanks to **git init**.

Section 2: Practice

Step-by-Step Guide to Using **git init**

1. Installing Git

Before using Git, ensure it is installed.

```
# Check if Git is installed
git --version

# If Git is not installed, install it (Ubuntu)
sudo apt update
sudo apt install git
```

Why is this important?

- Ensures Git is **installed and ready to use**.
 - Allows you to **track and manage versions**.
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2. Initializing a Git Repository

```
# Create a new project folder  
mkdir my_project  
cd my_project  
  
# Initialize Git in the folder  
git init
```

What happens here?

- A hidden **.git** folder is created inside **my_project**.
 - Git starts tracking files and changes.
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3. Checking if Git is Initialized

```
# Check the Git status  
git status
```

Expected Output (if no files are added yet):

```
On branch main  
No commits yet  
nothing to commit (create/copy files and use "git add" to track)
```

This means Git is **ready**, but no files are being tracked yet.

4. Adding and Committing Files to Git

```
# Create a new file  
echo "print('Hello, Git!')" > script.py
```



```
# Stage the file
git add script.py

# Commit the file with a message
git commit -m "Added script.py file"
```

What does this do?

- **git add** prepares the file for tracking.
 - **git commit** saves the file with a timestamp and message.
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5. Checking the **.git** Folder

If you list hidden files in your project, you will see **.git**:

```
ls -la
```

Output:

```
drwxr-xr-x  3 user user 4096 Feb 26 10:00 .git
```

Why is this important?

- The **.git** folder stores **all version control data**.
 - If deleted, the repository will **lose its version history**.
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Real-World Example

A **startup** is building a mobile app. Their **team of 5 developers** needs to:

1. **Track every change in the project.**
2. **Restore previous versions if needed.**
3. **Collaborate without overwriting each other's work.**



What did they do?

1. Ran **git init** in their project folder.
2. Used **git add** and **git commit** to track their changes.
3. Pushed their repository to **GitHub** for remote collaboration.

Result? Faster development, **better teamwork**, and **version control**.

Section 3: Know More

Frequently Asked Questions

1. **What does **git init** do?**

It **creates a new Git repository** in your project folder, allowing version control.

2. **Can I undo **git init**?**

Yes! Delete the **.git** folder:

```
rm -rf .git
```

3. This removes **all Git history and tracking**.

4. **How do I check if a directory is a Git repository?**

```
git status
```

5. If Git is initialized, it will show the **current branch and status**.

6. **Do I need to run **git init** for every project?**

Yes! Every new project must be initialized separately.

7. **Where can I practice Git commands?**

- Create a **GitHub account** and start a project.
 - Use **GitHub Desktop** for a graphical interface.
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Conclusion

The **git init** command is **the first step in using Git**. It enables developers to **track, manage, and collaborate on code efficiently**.

Start by **initializing a repository**, add files, and commit changes. Soon, you'll be **managing projects like a pro!**